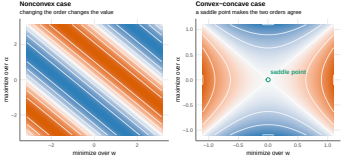
Optimization in Machine Learning

Constrained Optimization
Lagrangian Relaxation and Duality



Learning goals

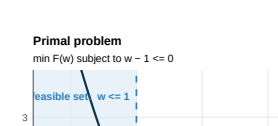
- Relax constraints with Lagrange multipliers
- Dual function, weak duality, and strong duality
- Min-max ordering, saddle points, and KKT conditions
- Example: deriving the hinge-loss SVM dual

WHY MOVE TO THE DUAL?

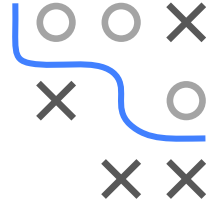
Primal problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^d} \quad F(\boldsymbol{x})$$
$$\text{s.t.} \quad g_i(\boldsymbol{x}) \leq 0, \quad i = 1, \ldots, m$$

- Primal methods enforce feasibility directly in the original variables.
- Dual methods **relax** the constraints and attach nonnegative multipliers to violations.
- Each fixed multiplier vector $\boldsymbol{\alpha}$ yields a **lower bound**; maximizing over $\boldsymbol{\alpha}$ tightens it.



LAGRANGIAN RELAXATION



- For inequality constraints, define the Lagrangian minimax objective

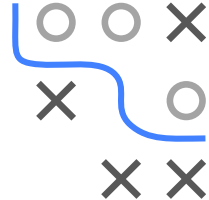
$$H(\boldsymbol{x}, \boldsymbol{\alpha}) := F(\boldsymbol{x}) + \sum_{i=1}^{m} \alpha_i g_i(\boldsymbol{x}), \qquad \boldsymbol{\alpha} \geq \mathbf{0}.$$

- The corresponding **dual function** is the relaxed optimum at fixed multipliers:

$$L(\boldsymbol{\alpha}) := \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}).$$

- The minimization is only with respect to $\boldsymbol{x}$; the multiplier vector $\boldsymbol{\alpha}$ is held fixed.
- Why $\alpha_i \geq 0$? If $g_i(\boldsymbol{x}) > 0$, then the violated constraint increases the objective by $\alpha_i g_i(\boldsymbol{x})$.
- If $g_i(\boldsymbol{x}) < 0$, the term can decrease the objective, which is why $L(\boldsymbol{\alpha})$ is a lower bound rather than a direct replacement for the primal.
- Equality constraints $h_j(\boldsymbol{x}) = 0$ are handled by adding $\sum_j \beta_j h_j(\boldsymbol{x})$ with sign-unrestricted multipliers $\beta_j \in \mathbb{R}$.

WEAK DUALITY



- For any feasible $\boldsymbol{x}$ and any multiplier vector $\boldsymbol{\alpha} \geq \mathbf{0}$,

$$H(\boldsymbol{x}, \boldsymbol{\alpha}) = F(\boldsymbol{x}) + \sum_{i=1}^{m} \alpha_i g_i(\boldsymbol{x}) \leq F(\boldsymbol{x}),$$

  because each $g_i(\boldsymbol{x}) \leq 0$ and each $\alpha_i \geq 0$.

- Taking the minimum over $\boldsymbol{x}$ on the left yields

$$L(\boldsymbol{\alpha}) = \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}) \leq F(\boldsymbol{x}) \qquad \text{for every feasible } \boldsymbol{x}.$$

- In particular, if p^* denotes the optimal primal value, then

$$L(\boldsymbol{\alpha}) \leq p^* \qquad \forall \boldsymbol{\alpha} \geq \mathbf{0}.$$

- Maximizing this lower bound gives the **dual problem**

$$d^* := \max_{\boldsymbol{\alpha} \geq \mathbf{0}} L(\boldsymbol{\alpha}) = \max_{\boldsymbol{\alpha} \geq \mathbf{0}} \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}),$$

  and therefore

$$d^* \leq p^*.$$

- This universal lower-bound property is called **weak duality**.

ORDERING MATTERS: MIN-MAX VS. MAX-MIN

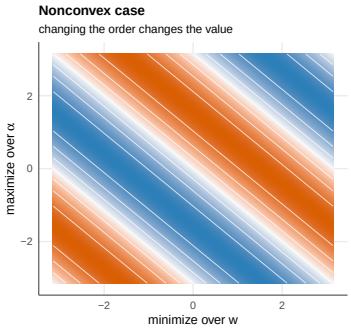
Dual:

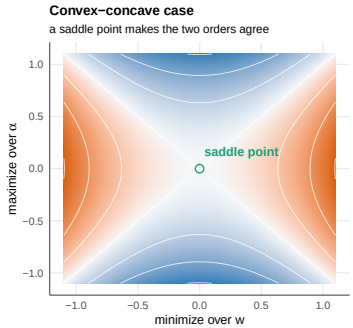
$$d^* = \max_{\boldsymbol{\alpha} \geq \mathbf{0}} \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha})$$

Primal:

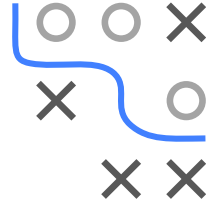
$$p^* = \min_{\boldsymbol{x}} \max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha})$$

- Always, $\max_{\boldsymbol{\alpha}} \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}) \leq \min_{\boldsymbol{x}} \max_{\boldsymbol{\alpha}} H(\boldsymbol{x}, \boldsymbol{\alpha})$.
- Without convex-concave structure, the order can change the value.
- If H is convex in $\boldsymbol{x}$ and concave in $\boldsymbol{\alpha}$, a saddle point makes the two orders equal.





THE PRIMAL IS THE REVERSED MINIMAX



- Fix the primal variable $\boldsymbol{x}$ and maximize $H(\boldsymbol{x}, \boldsymbol{\alpha})$ over $\boldsymbol{\alpha} \geq \mathbf{0}$.
- If $\boldsymbol{x}$ violates any constraint, say $g_k(\boldsymbol{x}) > 0$, then

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha}) = \infty,$$

  because we can send $\alpha_k \to \infty$.
- If $\boldsymbol{x}$ is feasible, then each penalty term satisfies $\alpha_i g_i(\boldsymbol{x}) \leq 0$, so the maximum is achieved at $\boldsymbol{\alpha} = \mathbf{0}$:

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha}) = F(\boldsymbol{x}).$$

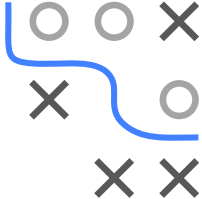
- Therefore,

$$\min_{\boldsymbol{x}} \max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha}) = \min_{\boldsymbol{x} \in \mathcal{S}} F(\boldsymbol{x}) = p^*.$$

- Primal and dual thus optimize the **same** Lagrangian objective, but in different orders:

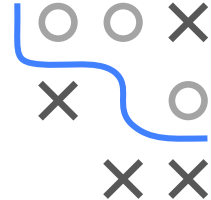
$$p^* = \min_{\boldsymbol{x}} \max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha}), \qquad d^* = \max_{\boldsymbol{\alpha} \geq \mathbf{0}} \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}).$$

STRONG DUALITY AND SLATER'S CONDITION



- Weak duality always gives
$$d^* \leq p^*.$$
- To obtain equality, we usually need convex structure:
  - $F(\boldsymbol{x})$ convex
  - each $g_i(\boldsymbol{x})$ convex
  - hence $H(\boldsymbol{x}, \boldsymbol{\alpha})$ convex in $\boldsymbol{x}$ and linear, therefore concave, in $\boldsymbol{\alpha}$
- A standard qualification is **Slater's condition**: there exists at least one strictly feasible point $\boldsymbol{x}_0$ such that
$$g_i(\boldsymbol{x}_0) < 0 \qquad \forall i = 1, \ldots, m.$$
- Under convexity plus Slater's condition, the duality gap disappears:
$$d^* = p^*.$$
- This is **strong duality**, and it is what makes dual optimization so useful in many machine-learning problems.

KKT CONDITIONS



- For differentiable convex problems under suitable regularity conditions, optimality is characterized by the **Karush-Kuhn-Tucker (KKT)** conditions.
- For inequality-constrained problems, a pair $(\boldsymbol{x}^*, \boldsymbol{\alpha}^*)$ is optimal if and only if:
- **Primal feasibility:**

$$g_i(\boldsymbol{x}^*) \leq 0 \qquad \forall i$$

- **Dual feasibility:**

$$\boldsymbol{\alpha}^* \geq \mathbf{0}$$

- **Complementary slackness:**

$$\alpha_i^* g_i(\boldsymbol{x}^*) = 0 \qquad \forall i$$

- **Stationarity:**

$$\nabla F(\boldsymbol{x}^*) + \sum_{i=1}^{m} \alpha_i^* \nabla g_i(\boldsymbol{x}^*) = \mathbf{0}$$

- Interpretation: inactive constraints have $\alpha_i^* = 0$, while active constraints can have nonzero multipliers.
- The multipliers act like **shadow prices**: they measure how strongly the objective pushes against each active constraint.

TOY EXAMPLE: ONE ACTIVE MULTIPLIER

Reuse

$$\min_{\boldsymbol{w}} F(\boldsymbol{w}) = (w_1 - 1.2)^2 + (w_1 - 1.2)(w_2 - 0.2) + (w_2 - 0.2)^2$$

$$\text{s.t.} \qquad 0 \le w_1 \le 1, \qquad 0 \le w_2 \le 0.9.$$

Write the four box constraints as

$$g_1(\boldsymbol{w}) = -w_1,\ g_2(\boldsymbol{w}) = w_1 - 1,\ g_3(\boldsymbol{w}) = -w_2,\ g_4(\boldsymbol{w}) = w_2 - 0.9.$$

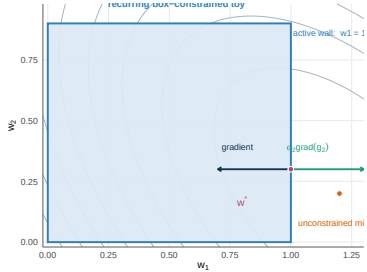
At the optimum $\boldsymbol{w}^* = (1, 0.3)^\top$,

$$\nabla F(\boldsymbol{w}^*) = (-0.3, 0)^\top.$$

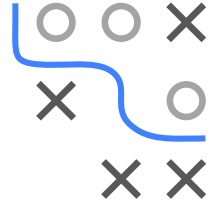
Hence the KKT multipliers are

$$\boldsymbol{\alpha}^* = (0, 0.3, 0, 0)^\top :$$

only the right wall is active, and its multiplier balances the push of the objective.



GENERAL RECIPE FOR DERIVING A DUAL

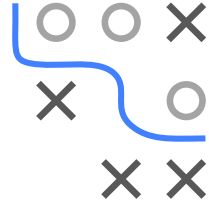


- **1. Start from a constrained primal.** Introduce slack variables if they simplify the objective or the constraints.
- **2. Form the Lagrangian** $H(\boldsymbol{x}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ by attaching a multiplier to each constraint.
- **3. Eliminate primal variables** by computing

$$L(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}, \boldsymbol{\beta}).$$

  Usually this is done via stationarity conditions with respect to the primal variables.
- **4. Substitute back** to obtain a pure optimization problem in the dual variables, together with any remaining sign or box constraints.
- **5. Solve the dual** and then recover the primal optimum from the stationarity relations.
- In many important examples, deriving the dual is mostly algebra: differentiate, substitute, simplify, and interpret the remaining constraints.

ML EXAMPLE: SVM PRIMAL AND STATIONARITY



- Start from the soft-margin SVM with slack variables:

$$\min_{\boldsymbol{w}, b, \boldsymbol{x}_1^{(i)}, \ldots, \boldsymbol{x}_n^{(i)}} \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C \sum_{i=1}^{n} \boldsymbol{x}_i^{(i)}$$

$$\text{s.t.} \qquad 1 - y^{(i)}\left(\boldsymbol{x}^{(i)\top} \boldsymbol{w} + b\right) - \boldsymbol{x}_i^{(i)} \leq 0, \qquad \boldsymbol{x}_i^{(i)} \geq 0.$$

- Introduce multipliers $\alpha_i \geq 0$ for the margin constraints and $\gamma_i \geq 0$ for the nonnegativity constraints.
- Stationarity with respect to the primal variables gives

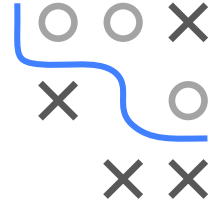
$$\boldsymbol{w} = \sum_{i=1}^{n} \alpha_i y^{(i)} \boldsymbol{x}^{(i)}, \qquad \sum_{i=1}^{n} \alpha_i y^{(i)} = 0, \qquad C - \alpha_i - \gamma_i = 0.$$

- The equality constraint comes from differentiating with respect to the intercept b.
- Since $\gamma_i \geq 0$, the dual variables satisfy the box constraints

$$0 \leq \alpha_i \leq C.$$

- Key interpretation: the optimal weight vector is a linear combination of training points.

ML EXAMPLE: RESULTING SVM DUAL



- Eliminating the primal variables leaves the dual quadratic program

$$\max_{\substack{0\le\alpha_i\le C\\ i=1,\dots,n}} \left[\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y^{(i)}y^{(j)}\boldsymbol{x}^{(i)\top}\boldsymbol{x}^{(j)}\right].$$

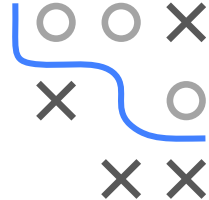
$$\text{s.t.} \qquad \sum_{i=1}^{n}\alpha_i y^{(i)} = 0.$$

- This is a **box-constrained QP plus one equality constraint** in the dual variables only.
- Once the optimal multipliers are known, the primal separator follows from

$$\boldsymbol{w}^* = \sum_{i=1}^{n}\alpha_i^* y^{(i)}\boldsymbol{x}^{(i)}.$$

- Only points with $\alpha_i^* > 0$ contribute to $\boldsymbol{w}^*$; these are the **support vectors**.
- The dual is especially attractive because it depends on the data only through dot products $\boldsymbol{x}^{(i)\top}\boldsymbol{x}^{(j)}$, which later enables kernel methods.

TAKE-AWAY



1. Lagrangian relaxation converts constraints into multiplier variables.
2. The dual value is a lower bound:

$$d^* = \max_{\boldsymbol{\alpha} \geq \mathbf{0}} \min_{\boldsymbol{x}} H(\boldsymbol{x}, \boldsymbol{\alpha}) \leq p^*.$$

3. The primal uses the same objective in reversed order:

$$p^* = \min_{\boldsymbol{x}} \max_{\boldsymbol{\alpha} \geq \mathbf{0}} H(\boldsymbol{x}, \boldsymbol{\alpha}).$$

4. For convex problems with Slater's condition, strong duality and the KKT conditions let us solve and certify optimality through the dual.

Dual methods are especially attractive when the Lagrangian lets us eliminate many primal variables and leaves a simpler problem in the multipliers.